Plan — First Learned s1 KINETIC Delivery

A causal, teacher-labelled coin-following torque policy over the frozen scaffold (K0–K4)

HyMeKo coin-delivery campaign (`recovery/coin-r9-causal-residual-delivery`)

2026-07-27

Abstract

The mechanical picture is settled and the K6 monitor is not the blocker. On dev cradle `s1` (seed 14250) the θ-primitive teacher delivers strict K6 (min $d_{tz} \approx 18.5$ mm; release at $d_{tz} \approx 23$ mm, $v_{\parallel} \approx 0.131$; passive coast into the zone). The frozen `APPROACH_MOMENTUM_BUILD` reaches teacher-close momentum ($v_{\parallel} \approx 0.316$ vs teacher 0.322); the `VELOCITY_FLOOR/KINETIC` mode transports the coin *while moving* (no stiction stall) from 75 mm to $\approx$48–51 mm; but neither hand-tuning nor a 6-param position-varying profile (216 CEM evals) holds the delicate light contact to the 20–30 mm release corridor. **The single missing skill is a learned, causal KINETIC torque policy that follows the moving coin in light contact to a close-and-moving release.** This plan builds it and only it, targeting exactly one success gate: `FIRST_LEARNED_S1_K6_DELIVERY`. No new atlas, no clustering, no diagnostic arc. *Deliver the coin.*

1 Scope and goal

Goal. A learned KINETIC coin-following policy, deployed with no teacher/oracle in the loop, delivers the `s1` coin into the strict K6 zone. **Architecture (frozen $\leftrightarrow$ learned boundary):**

frozen APPROACH → learned local KINETIC coin-following → G0 release guard → frozen passive coast → frozen K6.

Non-goals (explicit stop conditions). No return to hand-tuned transport profiles. No five-cradle atlas before the first learned s1 delivery (only what the s1 feedback bank strictly needs). The secondary local-policy atlas (LIGHT-slide {s1/14250, 17750}; FIRM-guided {s3/14750, 19500, 16500}) is deferred until after this gate.

2 The frozen contract (must hold at every step)

- Physics + motion contract unchanged (`env/motion_contract.py`, governed $\Delta\tau$, slew, joint/coin speed hard caps). All torque enters through the existing `_servo`→governor stack; no raw actuator writes.
- K6 monitor unchanged: `CENTER_TOL`$= 0.02$ m, `SETTLE_VEL`$= 0.06$ m/s, `HELD_DWELL`$= 6$ (`coin_rl_env.py:13--15`); authoritative verdict `delivery_success(m,cfg)` (`forward_displacement.py:232`).
- `s4`/`s7` validation-only; `f1--f4 SEALED_NOT_EVALUATED`; not opened before the first s1 delivery.
- No teleport, no coin-state edit, no hidden force, no illegal oracle deployment. The teacher is used *offline* to label; the deployed policy is the learned clone only.

3 Affected files

CORE.YAML items touched: none (verified against `CORE.YAML`: the coin RL work is entirely outside `hymeko_core/hymeko_query/hymeko_client/hymeko_daemon/parser` and touches no pinned dependency). All new code lives under `hymeko_rl/` (a non-core crate-adjacent Python package) and `tests//reports//docs/plans/` (allowlisted).

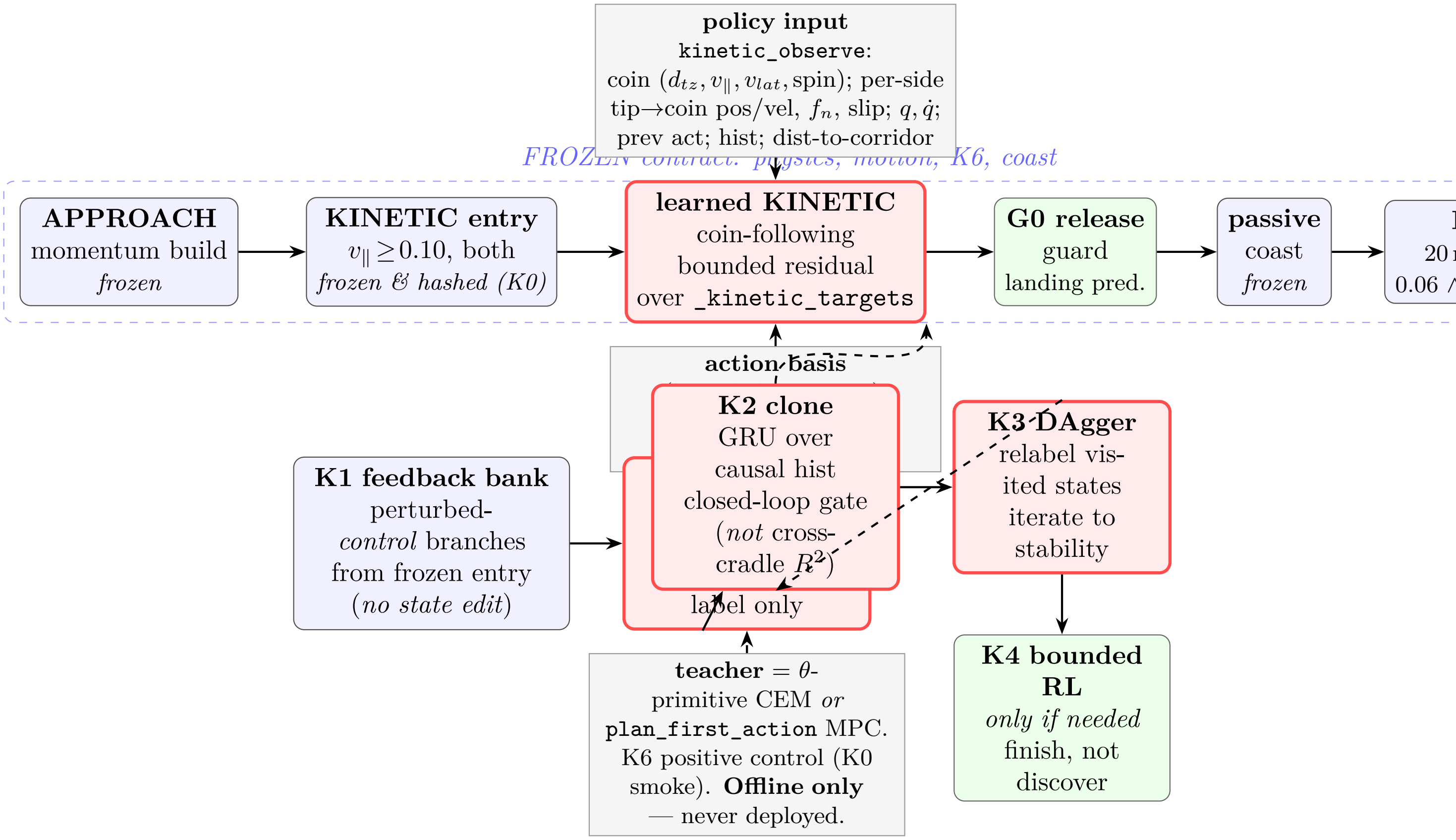

Figure 1: Deployment chain (top) and the offline machinery that produces the learned KINETIC block (bottom). The teacher is an *offline* label source, never deployed. Blue = frozen; red = learned; green = guard.

This change (K0) — new files only:

- `hymeko_rl/coin_delivery/theta_option/kinetic_contract.py` — the K0 module: `TransportSnapshot` (duck-typed, branchable mid-transport snapshot), `freeze_kinetic_entry` (deterministic frozen KINETIC-entry snapshot on s1), `kinetic_observe` (the *single* canonical feature extractor used by both bank-build and deploy), `receding_horizon_relabel` (teacher first-action label from a transport snapshot), `KineticEntry`/`TransportAdmissibility` dataclasses.
- `hymeko_rl/experiments/coin_kinetic_positive_control.py` — the K0 positive-control smoke: the two candidate teachers replanned from the frozen KINETIC entry → strict-K6 verdict on s1 → pick the relabeler.
- `hymeko_rl/tests/test_coin_kinetic_contract.py` — unit + integration tests.
- `reports/2026-07-27-coin-r9-learned-s1-kinetic-delivery-k0.md` — the K0 report.

Later milestones (planned, not built in K0): `kinetic_feedback_bank.py` (K1), `kinetic_clone.py` + a `KineticResidualController` in `hybrid_approach.py`'s residual style (K2), `kinetic_dagger.py` (K3), `landing_guard.py` (G0 predictor promoted to a fittable object), optional `kinetic_bounded_rl.py` (K4). **Read-only reuse:** `contact_velocity.py` (`CradleSnapshot.branch`), `forward_displacement.py` (`cem_search`, `rollout_primitive`, `delivery_success`), `coin_v3_receding_horizon.py` (`plan_first_action`, `_snapshot`/`_restore`), `theta_option/hybrid_a` (KINETIC phase), `theta_option/velocity_transport.py` (`velocity_rollout`), `theta_option/residual_adapter.py` (bounded-residual pattern), `theta_option/teacher_bank.py` (`load_harness`, `acquire_snapshot`).

4 Interface changes (signatures, contracts)

All additive; no existing signature changes. Key new contracts:

- `TransportSnapshot` — duck-types the subset of `CradleSnapshot` the teacher needs (`.branch()`→`deepcopy`, `.stack`, `.prev_tau`, `.lo`, `.hi`, `.q_hold`) built from an *arbitrary* mid-transport live `rl` *without* the B_v straddle gate. **Post:** `branch()` is bit-identical across calls (deterministic deepcopy); `admissibility()` reports (dual contact, straddle sign, motion-contract) as a *query*, never a silent raise.
- `kinetic_observe(rl, hist)` → `(feat: np.ndarray, hframe: list)` — extends the frozen `_kinetic_feats` (`coin_kinetic_bc.py:41`) with per-side tip→coin relative position/velocity, per-side slip, and distance-to-release-corridor. **Pre:** live `rl`. **Post:** identical output whether called in batch (bank build) or streaming (deploy) — the "batch = streaming" invariant; *no future/teacher/K6 leak*.
- `receding_horizon_relabel(tsnap, teacher, *, budget, warm)` → `RelabelResult` — runs the chosen receding-horizon teacher from `tsnap`, returns `(first_action, delivers_k6, teacher_provenance)`. **Contract:** only the FIRST action is exported as a label; the teacher's lookahead is baked into the label, never into the policy input. Deterministic given `(tsnap, budget, seed)`.
- `freeze_kinetic_entry(seed=14250, params)` → `KineticEntry` — rolls the frozen `HybridApproachController(kinet` via `velocity_rollout` until phase `APPROACH`→`KINETIC`, captures (q_{pos}, q_{vel},prev_tau, phase state) at that step, content-hashes it. **Post:** same seed/params ⇒ bit-identical hash.

5 Test strategy (per layer)

Unit (`test_coin_kinetic_contract.py`, `cargo`-fast, deterministic seeds): `TransportSnapshot.branch()` bit-identity across two calls; `kinetic_observe` batch=stream identity on a fixed rollout; `kinetic_observe` contains no future/teacher/K6 field (schema assertion + a leak probe: a shuffled future must not change the vector); `freeze_kinetic_entry` hash determinism; a first-action label is length-4 and governed-admissible. **Integration:** from the frozen KINETIC entry, one teacher replan produces a governed action; a short teacher-labelled closed segment stays within the motion contract. **Positive-control smoke (production scale, 1 seed):** the chosen teacher, replanned from the frozen s1 KINETIC entry, delivers strict K6 (`delivery_success` True). This is the load-bearing K0 measurement; if it fails, **HALT** and localise (feature/history, teacher instability, action basis, or optimisation blocker) — do *not* conclude the skill is unlearnable (the teacher is a physical positive control). **Coverage:** every new public/private function exercised by ≥ 1 new test; `kinetic_observe` and the relabeler get regression tests that would fail against a naive (leaky / non-deterministic) implementation.

6 Performance budget

Measured anchors (this host, Apple-Silicon, `.venv` py3.11.15, mujoco 3.10.0): one governed rollout ≈ 0.83 s at peak RSS 0.24 GB; heaviest existing coin tests peak ≈ 0.20 GB; a full delivery CEM (`DELIVERY_CFG`, pop 32 × iters 6) ≈ 38 s/cradle (scout: 911 s / 24). **K0 budget:** unit+integration suite < 120 s wall; positive-control smoke ≤ 5 min (one seed, one/both teachers, warm-started local CEM); **peak RSS < 1 GB** (hard cap 16 GB, never approached). **Downstream budget (declared now for K1/K3):** a warm-started local relabel (CEM pop 16 × iters 3, centred on the s1 canonical θ) targets ≈ 5–10 s/label; a ∼200-label bank ≈ 20–30 min, *checkpointed*; policy inference (small GRU) ≪ 1 ms/step. A relabel-cost measurement in K0 reconciles this before any multi-hour K1 run (per the > 2× wall-estimate halt rule).

7 Risk anticipation

Performance-contract preservation. The learned KINETIC block is a *bounded residual over* `_kinetic_targets` (the `ResidualTipAdapter` pattern): at action $a = 0$ it reproduces the hand-tuned KINETIC *bit-for-bit* (update-zero identity, asserted). It inherits every contract the KINETIC branch already enforces — `kinetic_vcap` on the joint-velocity reference, the light-grip `fn_min_safe` contact-risk fallback, `kinetic_max_steps` horizon guard, and the slew/governor clip in `_servo`. New action axes (L/R differential, contact-normal) are added *inside* the same `clip(·,±slew)`→`governor` path; they cannot exceed the torque/slew/joint-speed limits. The G0 release remains guarded by the existing `kinetic_release_hi`/`kinetic_release_vmin` condition (and, later, the fittable landing guard) — the policy shapes the transport, it does not own an unshielded release bit.

Worst-case input / production scale. Target dataset = dev `s1` (seed 14250), canonical $\theta = [0.0515, 0.2179, 0.0164, 12.8078, 9.2369, 3.0388]$, teacher min d_{tz} 18.5 mm. The worst-case runtime input is a KINETIC-transport state where the coin has *slid ahead of the tips* (the measured ≈48 mm contact-loss failure): `admissibility()` must flag it, the relabeler must reject states from which no delivering replan exists (logged rejection rate, no silent truncation), and the deployed policy's contact-risk fallback must hand to the frozen brake rather than chase with raw torque. The bank must *cover* these near-loss states (that is where the skill lives), which is why K1 branches perturbed *control* (not state edits) and K3 DAgger relabels the clone's own visited states.

Named latent-contract greps (done): `hybrid_approach.py` comments "KINETIC OWNS the release", "preempt LAUNCH/REACHABILITY", "release only from genuine forward motion" — the residual must not subvert these; the update-zero test + the frozen release condition enforce it.

8 Rollback path

K0 adds only new files; rollback = delete them (no existing behaviour changes; the default `HybridApproachController` still has `kinetic_transport=False`, update-zero). Each later milestone is a separate commit gated on its own test; a failed Kn reverts to K(n−1) without touching the frozen scaffold. The frozen KINETIC entry is content-hashed, so a regression is detectable by hash mismatch.

9 Milestone ladder (executable)

K0 — snapshot & expert contract *(this change)*
Freeze the KINETIC-entry state; build `TransportSnapshot` + `kinetic_observe` + `receding_horizon_relabel`; **verify the teacher is a K6 positive control from the entry** (the gate that de-risks everything). First-action labels only; batch=stream features; no future info in the actor input.

K1 — local feedback bank (small, step-wise)
Re-measure cost first: 8–16 representative relabels warm-started from the entry-delivering θ (squeeze ≈ 0) with a *box-wide* legal window; report cost + success-rate; reconcile the ≈17 s/label estimate (> 2× halt rule) *before* sizing the bank. Then step-wise: **K1-A** 32 labels (schema, diversity, replan success-rate, cost audit); **K1-B** 128 labels (first feedback-clone + closed-loop smoke); **K1-C** 256–512 *only if needed*, from DAgger-visited states (not more teacher-neighbour samples). Bank quality = *distinct physical states / successful replans / v_par–slip–force– geometry spread / first-action variance* — **not** sample count. Branch perturbed *control* (`v_floor`, `kinetic_squeeze`, L/R imbalance, prev-action noise, teacher-trace neighbourhoods) through real physics — *physically legal, snapshot-branched, not state-edited.* Relabel each admissible state (K0). Log rejection rate.

K2 — closed-loop clone
Small structured encoder + GRU over the causal history → the bounded KINETIC action basis, BC on the K1 bank. **Gate `LOCAL_KINETIC_FEEDBACK_SKILL_PASS` (not BC-loss / action-R^2):**

the actor, running *without the teacher*, from entry-neighbourhood perturbations stably holds-or-correctly-releases the moving coin, improves reach to the close-and-moving release manifold, does not re-clamp, and reaches strict s1 K6 at least once. (The prior BC's `CROSS_CRADLE_BC_UNDERPOWERED` was small-$N/2$-cradle; here it is single-cradle by design — cross-cradle R^2 is not the gate.)

K3 — DAgger / receding-horizon correction
: Roll the clone, relabel its own visited states with the teacher (first-action only, teacher never deployed), retrain until closed-loop delivery stabilises — the feedback data the prior BC lacked.

K4 — bounded RL (only if needed)
: A small bounded residual over the clone *only* if it reliably reaches the release corridor but the final K6 needs fine correction. RL finishes; it does not discover a missing primitive.

10 Gate ladder

1. `REPLAN_FROM_KINETIC_ENTRY_DELIVERS_S1` *(K0 — done)*: a legal causally-replanned torque primitive from the frozen entry reaches strict K6 (min d_{tz} 15.7 mm). The interface-side positive control for the learned skill — **not** a learned delivery; the teacher/CEM still runs the loop.
2. `LOCAL_KINETIC_FEEDBACK_SKILL_PASS` *(K2 — next real gate)*: the learned actor, *teacher-free*, holds-or-correctly-releases the moving coin over entry-neighbourhood perturbations, improves release-manifold reach, does not re-clamp, and reaches strict s1 K6 at least once. Not BC-loss / action-R^2.
3. `FIRST_LEARNED_S1_K6_DELIVERY` *(campaign gate)*: learned KINETIC policy the sole thing in the loop; no teacher/oracle in deployment; strict K6 **PASS**; safety **PASS**; physics + monitor unchanged; deterministic replay; full provenance; a video/GIF artifact; an event-aligned trace (APPROACH exit, KINETIC entry, contact-force profile, $v_\parallel$, release state, passive landing, K6 dwell).

11 Empty-plan-dir hygiene

If this plan is abandoned before K0 lands, delete `docs/plans/2026-07-27-coin-r9-learned-s1-kinetic-delivery/` in the same session — an empty/orphan plan dir reads as in-progress work and must not linger.